

Aleksei Gregory Sorokin

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Research Scientific Machine Learning, Gaussian Processes, Quasi-Monte Carlo, Probabilistic Numerics

Programming Python (PyTorch, GPyTorch, Pandas, Matplotlib), Julia, C, MATLAB, R, SQL, Wolfram

Tools AWS (SageMaker, EC2), GitHub (Actions, Pages), L^AT_EX, Docker

Education

2026.01 – 2028.05 **Postdoc in Statistics**. University of Chicago (UChicago). Advisors *Yuehaw Khoo* and *Lek-Heng Lim*.

2021.08 – 2025.12 **PhD in Applied Math**. Illinois Institute of Technology (IIT). GPA 3.89/4. Advisor *Fred J. Hickernell*.

2017.08 – 2021.05 **Master of Data Science**. IIT. Summa Cum Laude. GPA 3.94/4.

2017.08 – 2021.05 **B.S. in Applied Math, Minor in Computer Science**. IIT. Summa Cum Laude. GPA 3.94/4.

Experiences

2025.01 – 2025.12 **DOE SCGSR Fellow in Applied Math** at **Sandia National Laboratories**, Livermore, CA. I produced scientific ML models for machine-precision solutions to nonlinear PDEs [1]. I developed scalable multifidelity Gaussian process regression models and open-source software implementations [10, 11, 7].

2021.08 – 2025.01 **Teaching Assistant** at **IIT**. I led reviews for PhD qualifying exams in analysis and computational math.

2024.05 – 2024.08 **Scientific Machine Learning Researcher** at **FM (Factory Mutual Insurance Company)**. I deployed scientific ML models, including PINNs and DeepONets, to accelerate CFD fire simulations [13].

2023.05 – 2023.08 **Graduate Intern** at **Los Alamos National Laboratory**. I modeled multifidelity solutions to PDEs with random coefficients using efficient and error-aware Gaussian process regression models [14].

2022.09 – 2022.11 **Participant in Argonne National Laboratory's Course on AI-Driven Science on Supercomputers**. Key topics included handling large-scale data pipelines and parallel training for neural networks.

2022.05 – 2022.08 **Givens Associate Intern** at **Argonne National Laboratory**. I derived error bounds and a sequential sampling method for efficiently estimating failure probabilities with probabilistic models [6].

2021.05 – 2021.08 **ML Engineer Intern** at **SigOpt (acquired by Intel)**. On a six-person ML team, I contributed production code for meta-learning model-aware hyperparameter tuning via Bayesian optimization [18].

2018.05 – 2019.08 **Instructor** of the **Computer Discovery Program, STARS Computing Corps** at **IIT**. I taught and developed curriculum for middle school and high school girls to learn programmatic thinking in Python.

Open-Source Software

QMCPy Quasi-Monte Carlo Python Software (qmcsoftware.org/qmcpy). I led dozens of collaborators across academia and industry to develop QMC sequence generators, automatic variable transformations, adaptive error estimators, and diverse use cases [5, 8, 9, 12, 19, 15, 20, 3, 4].

FastGPs Scalable Gaussian Processes in Python (alegresor.github.io/fastgps). FastGPs pairs lattice/digital-net designs with matching shift-invariant kernels, reducing GP costs from $\mathcal{O}(n^3)$ time and $\mathcal{O}(n^2)$ storage to $\mathcal{O}(n \log n)$ time and $\mathcal{O}(n)$ storage. This supports GPU scaling, batched inference, hyperparameter optimization, multifidelity GPs, and efficient Bayesian cubature [11, 10, 7].

TorchOrthoPolys Orthogonal Polynomials in PyTorch (alegresor.github.io/TorchOrthoPolys) with GPU support.

QMCToolsCL Randomized Quasi-Monte Carlo Sequences in C/OpenCL (qmcsoftware.org/QMCToolsCL).

QMCGenerators.jl Randomized Quasi-Monte Carlo Sequences in Julia (alegresor.github.io/QMCGenerators.jl).

Awards

2025.01 – 2025.12 **DOE SCGSR Fellow in Applied Math**, Sandia National Laboratories, Livermore, CA.

2025.01 **Karl Menger Student Award for Exceptional Scholarship (Graduate)**, IIT.

2024.01 **College of Computing Excellence in Dissertation Research**, IIT.

2024 **Teaching Assistant Award**, IIT.

2023.08 **Outstanding Math Poster**, Los Alamos National Laboratory.

2017.08 – 2025.05 **Dean's List Member**, IIT, every semester.

References

PhD Advisor **Fred J. Hickernell** (hickernell@illinoistech.edu) Professor of Applied Math, IIT.

Fellowship Mentor **Pieterjan M. Robbe** (pmrobbe@sandia.gov) Senior Member of Technical Staff, Sandia National Labs.

Postdoc Mentor **Yuehaw Khoo** (ykhoo@uchicago.edu) Assistant Professor, Department of Statistics, UChicago.

Postdoc Mentor **Lek-Heng Lim** (lekheng@uchicago.edu) Professor, Department of Statistics, UChicago.

- [1] Aras Bacho, Aleksei G. Sorokin, Xianjin Yang, Théo Bourdais, Edoardo Calvella, Matthieu Darcy, Alexander W. Hsu, Bamdad Hosseini, and Houman Owhadi. "Operator learning at machine precision". In: *Journal of Computational Physics* 566 (2026), p. 115240. DOI: 10.1016/j.jcp.2026.115240.
- [2] Eda Gjergo, Zhi-Yu Zhang, Pavel Kroupa, Aleksei G. Sorokin, Zhiqiang Yan, Ziyi Guo, Tereza Jerabkova, Akram Hasani Zonoozi, and Hosein Haghi. "Massive Star Formation at Supersolar Metallicities: Constraints on the Initial Mass Function". In: *Research in Astronomy and Astrophysics* 26.2 (Jan. 2026), p. 025003. DOI: 10.1088/1674-4527/ae1f79.
- [3] Fred J. Hickernell, Nathan Kirk, and Aleksei G. Sorokin. "Quasi-Monte Carlo methods: what, why, and how?" In: *Monte Carlo and Quasi-Monte Carlo Methods 2024*. Ed. by Christiane Lemieux and Ben Feng. Springer, 2026, pp. 3–40. DOI: 10.1007/978-3-032-10590-5_1.
- [4] Aadit Jain, Fred J. Hickernell, Art B. Owen, and Aleksei G. Sorokin. "Empirical Bernstein and betting confidence intervals for randomized quasi-Monte Carlo". In: *Information and Inference: A Journal of the IMA* 15.1 (Mar. 2026), iaag003. DOI: 10.1093/imaiai/iaag003.
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- [6] Aleksei G. Sorokin and Vishwas Rao. *Credible intervals for probability of failure with Gaussian processes*. 2026. DOI: 10.48550/arXiv.2311.07733.
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- [10] Aleksei G. Sorokin, Pieterjan Robbe, Gianluca Geraci, Michael S. Eldred, and Fred J. Hickernell. *Fast Bayesian multilevel quasi-Monte Carlo*. 2025. DOI: 10.48550/arXiv.2510.24604.
- [11] Aleksei G. Sorokin, Pieterjan Robbe, and Fred J. Hickernell. "Fast Gaussian process regression for high dimensional functions with derivative information". In: *Proceedings of the First International Conference on Probabilistic Numerics*. Ed. by Motonobu Kanagawa, Jon Cockayne, Alexandra Gessner, and Philipp Hennig. Vol. 271. Proceedings of Machine Learning Research. PMLR, 2025, pp. 35–49. URL: <https://proceedings.mlr.press/v271/sorokin25a.html>.
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